



Copper

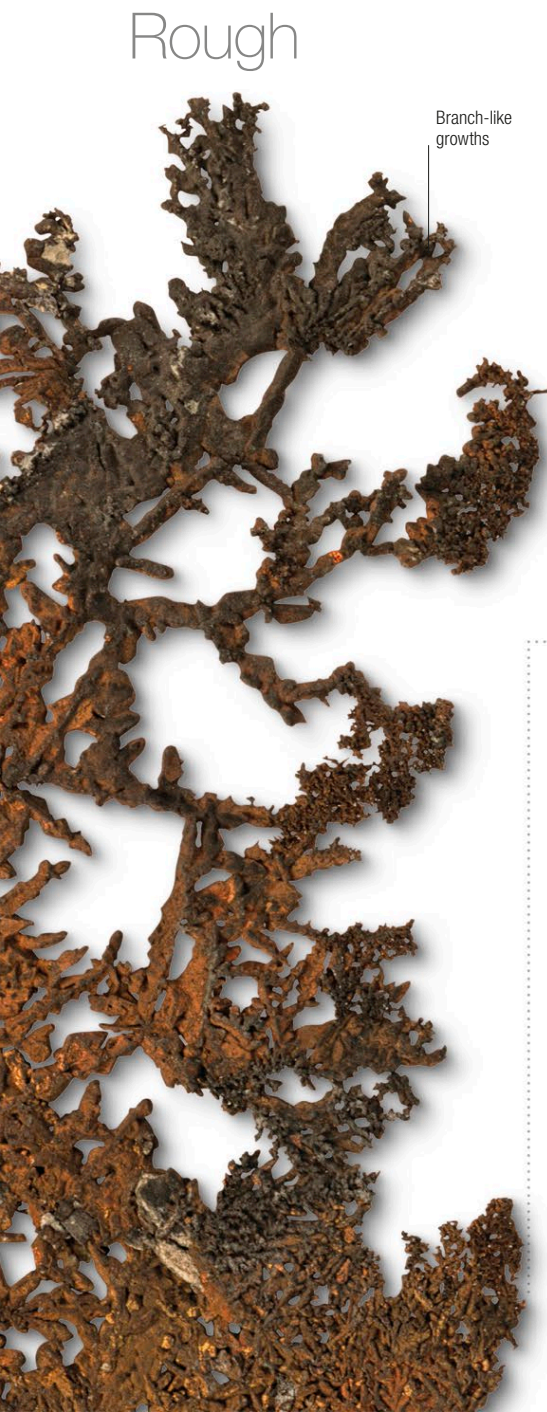
△ Sample of the mineral bornite, a principal ore of copper

Copper was the first metal to be used by humans – it naturally occurs in its pure form, and was used in early casting and decorative arts; it also forms a part of bronze, which was the first purposefully made alloy. Copper jewellery has been worn for millennia, and is popular in alternative medicine. It is also an extremely effective electrical conductor: this property, and the fact that it is extremely resistant to corrosion, have made copper the most widely used material for electrical wiring in the modern age.

Specification

Chemical name Copper | **Formula** Cu | **Colours** Orange-red | **Crystal system** Cubic | **Hardness** 2.5–3.0
SG 8.89 | **RI** 2.43 | **Lustre** Metallic | **Streak** n/a
Locations Chile, USA, Indonesia

Rough



Branch-like growths



Dendritic copper | The most dramatic form of native copper, dendritic copper occurs when copper crystals form thin, branching, fern-like sheets such as this.

Copper sheets



Native copper | In this mixed rocky specimen, thin, leaf-like sheets of native copper are interlayered with a piece of quartz groundmass.

Cast



Cast sphere | This bead of pure copper has been cast in the form of a sphere. Material such as this would be suitable for smelting or re-casting.



Four-sided jar | This bronze, four-sided *fanghu*, or square jar, originates from China c.475–221 BCE. It was decorated with green malachite, most of which is now lost.

Copper jewellery is said to protect against health problems

Garnet "eye"



Copper alloy brooch | This Anglo-Saxon bird brooch made of a copper alloy was found at Bekesbourne Anglo-Saxon cemetery, UK. It originates c.5th–8th century CE.